AME 5102: Applied Probability and Statistics

(L-T-P-C: 3-0-0-3)

Total hours: 36 hours

Course Outcome

CO1: Construct Bayesian models for quantifying uncertainty in practical problems.

CO2: Model random phenomena using random variables.

CO3: Use sample information and perform hypothesis-test analysis using an appropriate statistical technique to explain attributes of a population.

Unit I **No. of hours: 10 hrs**

Counting; Probability Concepts; Conditional Probability

Multiplication rule; permutation; combination - Sampling: with/without replacement and order matters/does not matter - Binomial & multinomial coefficients - Distribution problems; Set theory; sample space; outcomes; events - Frequency based definition of probability - Equally likely vs. not equally likely outcomes - Axioms of probability.

Conditional probability; Decomposition and the law of total probability - Bayes' rule - intuition, dependence/independence of events – Error metrics: sensitivity, specificity, and precision; ROC and Precision-Recall curves.

Unit II **No. of hours: 14 hrs**

Random Variables

Modelling using discrete random variables: Bernoulli, geometric, binomial, negative binomial, hypergeometric, and Poisson distributions - Probability mass function and cumulative distribution function - Expectation and variance: discrete case - Modelling using continuous random variables: uniform, normal, and exponential distributions; probability density function - Expectation and variance: continuous case.

Unit III **No. of hours: 12 hrs**

Sampling and Parameter Estimation

Population and sample - Statistic & sampling distribution - Sample mean and variance - Central limit theorem – intuition and applications.

Point estimation - Standard error - Interval estimation: interpretation of confidence interval - Hypothesis testing: p-values, significance level and their interpretations, application to analysis of one- /two-sample mean and paired data.

A/B testing and applications.

AME 5152: Applied Probability and Statistics Lab

(L-T-P-C: 0-0-3-1)

Total hours: 36 hours

Course Outcome

CO1: Understand probability concepts through visualization and frequency-based interpretations.

CO2: Construct models for discrete and continuous random phenomena using simulations.

CO3: Design and apply hypothesis tests followed by interpretation of results.

Unit I **No. of hours: 9 hrs**

Counting; Probability Concepts; Conditional Probability

Understand the basic principles of the R programming language; Develop short code snippets to understand the basic principles of sampling and probability; Visualise and interpret probability concepts through a frequency-based approach; Program and analyse Bayesian models for practical problems; Interpret error metrics such as sensitivity, specificity, & precision, and plot and interpret ROC and Precision-Recall curves for practical problems.

Unit II **No. of hours: 15 hrs**

Random Variables

Understand and apply R functions to simulate discrete and continuous random variables; Using sampling, compute and interpret different attributes of random variables; Visualise and interpret histograms and probability mass/density functions of random variables using state of the art visualisation libraries in R; Develop codes to model random phenomena using appropriate random variables.

Unit III **No. of hours: 12 hrs**

Sampling and Parameter Estimation

Visualise sample data through histograms; Compute estimates of population parameters using samples and communicate the uncertainty in the estimates; Use R in-built functions for performing hypothesis tests; Interpret and communicate the results of hypothesis tests.

References

1. Digital Dice: Computational Solutions to Practical Probability Problems, Paul Nahin, Princeton University Press
2. Applied Data Science with R Specialization from IBM (Courses 1, 3, and 4, https://www.coursera.org/specializations/applied-data-science-r)

3. Introduction to Probability, Charles M. Grinstead, American Mathematical Society; 2nd Revised Edition 1997. Available online at https://chance.dartmouth.edu/teaching_aids/books_articles/probability_book/amsbook.mac.pdf
4. A First Course in Probability, Sheldon Ross, 9th Edition, Pearson Education India; 9th Edition, 2013
5. Statistics without Tears: An Introduction for Non-Mathematicians (Paperback), Derek Rowntree, Penguin UK
6. Biostatistics Open Learning textbook – Online resource from University of Florida available at https://bolt.mph.ufl.edu/6050-6052/
7. All of Statistics: A Concise Course in Statistical Inference, Larry Wasserman – Springer